

Math 110 – Abstract Linear Algebra

Comprehensive exercise solutions

Axler, *Linear Algebra Done Right* (4th ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Adding a solution

To record a solved exercise, type under its chapter banner:

```
\exercise{2A.5} \begin{solution} ... \end{solution}
```

Restate the problem first with `\begin{problem}[title]... \end{problem}`; structure proofs with `claim`, `idea`, `proofsketch`.

Contents

Chapter 1	Vector Spaces	2
Chapter 2	Finite-Dimensional Vector Spaces	3
Chapter 3	Linear Maps	4
Chapter 4	Polynomials	5
Chapter 5	Eigenvalues and Eigenvectors	6
Chapter 6	Inner Product Spaces	7
Chapter 7	Operators on Inner Product Spaces	8
Chapter 8	Operators on Complex Vector Spaces	9
Chapter 9	Multilinear Algebra and Determinants	10

Chapter 1 Vector Spaces

Axler, 1A–1C: $\mathbb{R}^n$ and $\mathbb{C}^n$; definition of vector space; subspaces

Chapter 2 Finite-Dimensional Vector Spaces

Axler, 2A–2C: span and linear independence; bases; dimension

Chapter 3 Linear Maps

Axler, 3A–3F: the vector space of linear maps; null spaces and ranges; matrices; invertibility; products and quotients; duality

Chapter 4 Polynomials

Axler: zeros, division algorithm, factorization over $\mathbb{C}$ and $\mathbb{R}$

Chapter 5 Eigenvalues and Eigenvectors

Axler, 5A–5E: invariant subspaces; minimal polynomial; upper-triangular matrices; diagonalizable operators; commuting operators

Chapter 6 Inner Product Spaces

Axler, 6A–6C: inner products and norms; orthonormal bases and Gram–Schmidt; orthogonal complements and minimization

Chapter 7 Operators on Inner Product Spaces

Axler, 7A–7F: self-adjoint and normal operators; spectral theorem; positive operators; isometries and unitary operators; SVD

Chapter 8 Operators on Complex Vector Spaces

Axler, 8A–8D: generalized eigenvectors and nilpotent operators; generalized eigenspace decomposition; consequences; trace

Chapter 9 Multilinear Algebra and Determinants

Axler, 9A–9D: bilinear and quadratic forms; alternating multilinear forms; determinants; tensor products